Formulary: Investment and Financial Management

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Note
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1 Interest Calculation

C_0	Initial Capital (Present Value)
C_N	Final Capital (Future Value)
N	Number of Total Interest Payments
n	Maturity in Years
m	Number of Interest Payments per Year
r	Interest Rate (nominal)
r_{eff}	Effective Interest Rate
f	Fraction of predetermined period

Simple Interest

$$C_N = C_0 \cdot (1 + N \cdot r)$$

$$\mathrm{C}_f = C_0 \cdot (1 + f \cdot r)$$

Compound Interest

$$C_N = C_0 \cdot (1 + r)^N$$

Intra-Year Interest

$$N = m \cdot n$$

$$r_{eff} = \left(1 + \frac{r}{m}\right)^m - 1$$

$$\mathrm{C}_n = C_0 \cdot (1 + r_{eff})^n$$

Continuous Interest

$$C_N = C_0 \cdot e^{r \cdot n}$$

$$r_{eff} = e^r - 1$$

2 Annuity Calculation

C	Annuity-immediate; C_{due} Annuity-due
N	Maturity
r	Interest Rate
q	(1+r)
w	Annual growth rate of annuity, given geometrically growing annuity
g	(1+w)
d	Increase in annuity, given arithmetically growing annuity
FV	Final Value (annuity-immediate); FV_{due} (annuity-due)
PV	Present Value (annuity-immediate); PV_{due} (annuity-due)
C'	Fictitious periodic annuity (Replacement annuity)
m	Number of Interest Payments per Year
k	Time index

Constant Annuity

	annuity-immediate	annuity-due
Future Value	$FV = C \cdot \frac{q^N - 1}{q - 1}$	$FV_{due} = C_{due} \cdot q \cdot \frac{q^N - 1}{q - 1}$
Present Value	$PV = C \cdot \frac{1}{q^N} \cdot \frac{q^N - 1}{q - 1}$	$PV_{due} = C_{due} \cdot \frac{1}{q^{N-1}} \cdot \frac{q^N - 1}{q - 1}$

Arithmetically Growing Annuity

Future Value (annuity-immediate)
$$FV = \left(C + \frac{d}{q-1}\right) \cdot \frac{q^N - 1}{q - 1} - \frac{N \cdot d}{q - 1}$$

Future Value (annuity-due)
$$FV_{due} = \left(C_{due} + \frac{d}{q-1}\right) \cdot \frac{q \cdot (q^N - 1)}{q - 1} - \frac{N \cdot d \cdot q}{q - 1}$$

Present Value (annuity-immediate)
$$PV = \left(C + \frac{d}{q-1}\right) \cdot \frac{q^N - 1}{q^N \cdot (q - 1)} - \frac{N \cdot d}{q^N \cdot (q - 1)}$$

Present Value (annuity-due)
$$PV_{due} = \left(C_{due} + \frac{d}{q-1}\right) \cdot \frac{q^N - 1}{q^{N-1} \cdot (q - 1)} - \frac{N \cdot d}{q^{N-1} \cdot (q - 1)}$$

Initial Annuity C (annuity-immediate)
$$C = \left(FV + \frac{N \cdot d}{q-1}\right) \cdot \frac{q - 1}{q^N - 1} - \frac{d}{q - 1}$$
$$C_k = C + (k-1) \cdot d$$

Initial Annuity C (annuity-due)
$$C = \left(FV + \frac{N \cdot d \cdot q}{q-1}\right) \cdot \frac{q - 1}{q \cdot (q^N - 1)} - \frac{d}{q - 1}$$
$$C_{due,k} = C_{due} + (k-1) \cdot d$$

Geometrically Growing Annuity

It holds: $g = 1 + w$ and $q = 1 + r$

Future Value (annuity-immediate) $$FV = \begin{cases} C \cdot N \cdot q^{N-1} & \text{for } g = q \\ C \cdot \frac{g^N - q^N}{g-q} & \text{for } g \neq q \end{cases}$$

Future Value (annuity-due) $$FV_{due} = \begin{cases} C_{due} \cdot N \cdot q^{N} & \text{for } g = q \\ C_{due} \cdot q \cdot \frac{g^N - q^N}{g-q} & \text{for } g \neq q \end{cases}$$

Present Value (annuity-immediate) $$PV = \begin{cases} \frac{C \cdot N}{q} & \text{for } g = q \\ C \cdot \frac{(\frac{g}{q})^N - 1}{g-q} & \text{for } g \neq q \quad N = \dfrac{\ln\left(1 + \frac{PV \cdot (g-q)}{C}\right)}{\ln\left(\frac{g}{q}\right)} \end{cases}$$

Present Value (annuity-due) $$PV_{due} = \begin{cases} C_{due} \cdot N & \text{for } g = q \\ C_{due} \cdot q \cdot \frac{(\frac{g}{q})^N - 1}{g-q} & \text{for } g \neq q \quad N = \dfrac{\ln\left(1 + \frac{PV_{due} \cdot (g-q)}{C_{due} \cdot q}\right)}{\ln\left(\frac{g}{q}\right)} \end{cases}$$

Perpetual Annuity

	annuity-immediate	annuity-due
Present Value	$PV = \dfrac{C}{r}$	$PV_{due} = q \cdot \dfrac{C_{due}}{r}$

Growing Perpetual Annuity

It holds: $w < r$

	annuity-immediate	annuity-due
Present Value	$PV = \dfrac{C}{q-g} = \dfrac{C}{r-w}$	$PV_{due} = q \cdot \dfrac{C_{due}}{q-g} = q \cdot \dfrac{C_{due}}{r-w}$
Initial Annuity C	$C = \begin{cases} \dfrac{PV \cdot q}{N} & \text{for } g = q \\ PV \cdot \dfrac{g-q}{(\frac{g}{q})^N - 1} & \text{for } g \neq q \end{cases}$	$C = \begin{cases} \dfrac{PV_{due}}{N} & \text{for } g = q \\ PV_{due} \cdot \dfrac{g-q}{q \cdot ((\frac{g}{q})^N - 1)} & \text{for } g \neq q \end{cases}$
		$C_{due,k} = C_{due} \cdot g^{(k-1)}$

Fictitious periodic annuity
(Replacement Annuity)

	annuity-immediate	annuity-due
Fictitious periodic annuity	$C' = C \cdot \left[m + \frac{r \cdot (m-1)}{2}\right]$	$C'_{due} = C_{due} \cdot \left[m + \frac{r \cdot (m+1)}{2}\right]$

2.1 Repayment Calculation

A_k	Annuity payment in period k
N	Maturity
k	Foregone Periods
I_k	Interest Payment in Period k
T_k	Redemption Payment in Period k
D_0	Initial Debt
D_k	Debt in Period k
q	1+r

General Formulas

$$A_k = T_k + I_k$$
$$D_k = D_{k-1} - T_k$$

$$D_0 = \sum_{k=1}^{N} T_k$$

$$I_k = r \cdot D_{k-1}$$

Installment Repayment

$$A_k = D_0 \cdot (r \cdot (1 - \frac{k-1}{N}) + \frac{1}{N})$$

$$T_k = T = \frac{D_0}{N}$$

$$D_k = D_0 \cdot (1 - \frac{k}{N})$$

Annuity Repayment

$$A = D_0 \cdot \frac{q^N (q-1)}{q^N - 1} \qquad N = \frac{-\ln\left(1 - \frac{D_0 \cdot (q-1)}{A}\right)}{\ln(q)}$$

$$T_k = A \cdot \frac{1}{q^{N-K+1}}$$

$$D_k = D_0 \cdot \frac{q^N - q^k}{q^N - 1}$$

2.2 Bond Valuation

B_0	Present Value of a Coupon Bond
B_0^{ZB}	Present Value of a Zero Bond
B_N	Repayment (Face Value)
C	Coupon
N	Maturity
r	Interest Rate
D	Duration
D_{mod}	Modified Duration
I_{t_1,t_2}	Spot Rate from t_1 to t_2
r_{t_1,t_2}	Forward Rate from t_1 to t_2
k	Time index
T	End time
S	Intermediate point

Zerobond $$B_0^{ZB} = \frac{B_N}{(1+r)^N}$$

Coupon Bond $$B_0 = \left[C \cdot \frac{(1+r)^N - 1}{r} + B_N\right] \cdot \frac{1}{(1+r)^N}$$

Duration $$D = \frac{1}{B_0} \cdot \left[\sum_{k=1}^{N} \frac{k \cdot C_k}{(1+r)^k} + \frac{N \cdot B_N}{(1+r)^N}\right]$$

Duration of a portfolio $$\mathrm{D}_{pf} = \sum_{i=1}^{N} X_i \cdot D_i$$

Modified Duration $$D_{mod} = \frac{1}{1+r} \cdot D$$

Approximation of the Change in Present Value $$\frac{\Delta B_0}{B_0} \approx -D_{mod} \cdot \Delta r$$

Term Structure $$(1 + I_{t,T})^{T-t} = (1 + I_{t,S})^{S-t} \cdot (1 + r_{S,T})^{T-S}$$

Spot rate $$I_T = \sqrt[T]{\frac{\text{Face value}}{\text{Price}}} - 1$$

Face value $$= \text{Price} \cdot (1 + I_T)^T$$

2.3 Stock Valuation

P_t	Price of the Stock at time t
D_t	Dividend at time t
EPS	Earnings per Share
r_e	Equity Cost of Capital
P/E	Price/Earnings-Ratio
P/B	Price-to-Book Ratio (Market-to-Book Ratio)
V_E	Book Value of Equity
w	Growth Rate
p	Payout Ratio
ROE	Return on Equity
C	Coupon/Nominal Interest Rate on the Bond
t	Time
N	Maturity

Total Return
(One Year Horizon)

$$r_e = \frac{D_1 + P_1}{P_0} - 1 = \frac{D_1}{P_0} + \frac{P_1 - P_0}{P_0}$$

Dividend Discount Model

$$P_0 = \sum_{t=1}^{\infty} \frac{D_t}{(1 + r_e)^t}$$

Constant Growth

$$P_0 = \frac{D_1}{r_e - w} \qquad w = (1 - p) \cdot ROE$$

Changing Growth

$$P_0 = \sum_{t=1}^{N} \frac{D_0 \cdot (1 + w_a)^t}{(1 + r_e)^t} + \frac{D_1 \cdot (1 + w_a)^{N-1} \cdot (1 + w_b)}{r_e - w_b} \cdot \frac{1}{(1 + r_e)^N}$$

Multiples

$$P/E = \frac{P_0}{EPS_1} \quad P/B = \frac{P_0}{V_e}$$

PVGO (Present Value of Growth Opportunities) $= P_0 - P_0^* = P_0 - \frac{EPS_1}{r_e}$

$$\frac{P}{E} = \frac{1}{r_e} \cdot \left(1 + \frac{PVGO}{\frac{EPS_1}{r_e}}\right)$$

2.4 Option Valuation

C	Value of a Call Option
P	Value of a Put Option
S	Stock Price
K	Strike Price
q	Discount Factor
T	Time to expiration
r_f	Risk-free Interest Rate
T	Maturity
u	Share price factor for an upward movement
d	Share price factor in the event of a downward movement
p	Probability of Increase (risk-neutral)
$1-p$	Probability of Decrease (risk-neutral)

Binomialmodel

Hedge Ratio

$$m = \frac{uS - dS}{C_u - C_d}$$

Probability of Increase (risk-neutral)

$$p = \frac{(1 + r_f) - d}{u - d}$$

Value of a Call Option

$$C = \frac{p\, C_u + (1 - p)\, C_d}{1 + r_f}$$

Value of a Put Option

$$P = \frac{p\, P_u + (1 - p)\, P_d}{1 + r_f}$$

Put-Call Parity

$$S_0 + P_0 = Kq^{-T} + C_0$$

3 Financial Analysis

3.1 Definition of the Free Cash Flow to the Firm

	Earnings before Interest and Taxes	EBIT
−	Adjusted Tax Expense	$\tau_C \cdot$ EBIT
+	Depreciation and amortization	Depr
+	Other Non-Cash-Expenses	NCExp
−	Other Non-Cash-Earnings	NCEarn
±	Change in Net Working Capital	Δ NWC
=	Cash Flow from Operations	OCF
+	Cash Flow from Investments	CFI
=	Free Cash Flow to the Firm	FCF

3.2 Definition of the Free Cash Flow to Equity

	Net Income	NI
+	Depreciation and Amortization	Depr
+	Other Non-Cash Expenses	NCExp
−	Other Non-Cash Earnings	NCEarn
±	Change in Net Working Capital	Δ NWC
=	Net Cash Flow from Operating Activities	NCF
+	Cash Flow from Investments	CFI
+	Cash Flow from Financing Activities	CFF
=	Free Cash Flow to Equity	FCFE

3.3 The most important financial statement ratios

Invested Capital (IC)	Book Value of Equity + Net Debt
Net Debt (NFO)	Debt - Financial Assets
Debt-to-Capital Ratio	D/(E+D)
Debt-Equity Ratio (L)	D/E
Net Financial Leverage (NFL)	NFO/E
Debt-to-EV Ratio	NFO/(MV Equity+NFO)
Interest Coverage Ratio	EBIT (EBITDA) / Interest Expense
Current Ratio	Current Assets/Current Liabilities
Quick Ratio	(Cash + short-term investments + A/R) / Current liabilities
Return on Equity (ROE)	NI/E
Return on Invested Capital (ROIC)	EBIT(1-Tax rate)/IC
Net Financial Expense (NFE)	FE/NFO
EBIT Margin	EBIT/Sales
Net Profit Margin	NI/Sales
Earnings per Share (EPS)	NI/number of shares outstanding
Diluted Earnings per Share (dEPS)	NI/(number of shares outstanding · dilution factor)
Price-Earnings-Ratio (P/E)	Stock price/EPS = market capitalization/NI
EBITDA Multiple	EV/EBITDA
Market to Book Ratio	Stock price · number of shares outstanding / book value of equity

3.4 Diluted Earnings

EPS	earnings per share
a	number of outstanding shares
X	issue (exercise) price
S	current stock price
NV	nominal value of convertible bond
c	coupon rate
NI	net income
τ_C	Tax rate

$EPS(diluted) = \frac{EPS}{DF1 \cdot DF2}$

$DF1 = max[1; 1 + \frac{n}{a}(1 - \frac{X}{S})]$

$DF2 = max[1; \frac{1+n/a}{1+\frac{NV(1-\tau)c}{NI}}]$

3.5 Breakdown-Analyse

ROE	operating Return on Equity
$ROIC^{aT}$	Return on Invested Capital after Tax
NFL	Net Financial Leverage
NFE	Net Financial Expense
NFO	Net Debt (=Net Financial Obligation)
τ_C	Tax rate

DuPont Identity

$$ROE = Net\ Profit\ Margin \cdot Asset\ Turnover \cdot Equity\ Multiplier =$$

$$= \frac{\text{Net Income}}{Sales} \cdot \frac{Sales}{\text{Total Assets}} \cdot \frac{\text{Total Assets}}{Book\ Value\ of\ Equity}$$

Book-Leverage Equation

$$ROE = ROIC^{aT} + NFL\left[ROIC^{aT} - \frac{NFE \cdot (1-\tau_C)}{NFO}\right]$$

4 Investment Analysis

NPV	Net Present Value
t	Time index
T	Number of periods
t^*	Payback period
CF_t	Net payment in period n
r	Discount rate
IRR	Internal rate of return
τ_C	Tax rate

Net Present Value $$NPV = \sum_{t=0}^{T} \frac{CF_t}{(1+r)^t}$$

Internal Rate of Return $$\sum_{t=0}^{T} \frac{CF_t}{(1+IRR)^t} \stackrel{!}{=} 0$$

Payback Rule: Search for smallest

$t^* \leq T$, with: $$\sum_{t=0}^{t^*} CF_t \stackrel{!}{\geq} 0$$

Profitability Index $$PI = \frac{NPV}{Resources\ consumed}$$

Economic Value Added (EVA) $$EVA_t = EBIT_t \cdot (1 - \tau_C) - r \cdot IC_{t-1}$$

5 Capital increase

P_{cum}	Price of the old share (Price cum)
IP	Subscription price of the new shares
BV	Subscription ratio (old shares / new shares)
Div_a	Dividend old share
Div_n	Dividend new share
SR	Value of the subscription right
P_{ex}	Share price immediately after the capital increase
t	Years until the next dividend payment
q	$1 + r_E = 1 +$ equity cost of capital

5.1 Capital increase against cash contributions

Calculated price ex

$$P_{ex} = \frac{BV \cdot P_{cum} + IP}{BV + 1}$$

Value of the subscription right

$$SR = \frac{P_{cum} - IP}{BV + 1}$$

Value of the subscription right
with dividend disadvantage
for the new shares

$$SR = \frac{P_{cum} - IP - (Div_a - Div_n) \cdot q^{-t}}{BV + 1}$$

5.2 Capital increase from company funds

Calculated price ex

$$P_{ex} = \frac{BV \cdot P_{cum}}{BV + 1} = \frac{P_{cum}}{1 + \frac{1}{BV}}$$

6 Cost of Capital and CAPM

E, r_E	Value of equity and equity cost of capital
D, r_D	Value of debt and debt cost of capital
r_U	Cost of capital of the unlevered firm
r_{WACC}	Cost of capital of the levered firm
r_{RF}	Risk free rate
$\overline{r}_i$	Expected return of the asset i
$\overline{r}_{Mkt}$	Expected return of the market portfolio
β_i	Beta factor of the asset i
β_E	Beta factor of equity (equity beta)
β_D	Beta factor of debt (debt beta)
β_U	Beta factor of assets (asset beta/unlevered beta)
V^U, V^L	Value of the unlevered/levered firm

CAPM $\quad \overline{r}_i = r_{RF} + (\overline{r}_{Mkt} - r_{RF})\beta_i$

6.1 Cost of capital without taxes

Cost of capital of the unlevered firm $\quad r_U = \frac{E}{E+D} r_E + \frac{D}{E+D} r_D$

Equity cost of capital of the levered firm $\quad r_E = r_U + \frac{D}{E}(r_U - r_D)$

Unlevered beta $\quad \beta_U = \frac{E}{E+D}\beta_E + \frac{D}{E+D}\beta_D$

Equity beta $\quad \beta_E = \beta_U + \frac{D}{E}(\beta_U - \beta_D)$

6.2 Cost of capital with taxes

Weighted Average Cost of Capital $\quad r_{WACC} = \frac{E}{E+D} r_E + \frac{D}{E+D} r_D (1 - \tau_C)$

WACC with target leverage ratio $\quad r_{WACC} = r_U - \frac{D}{E+D}\tau_C r_D$

Unlevered beta (Hamada equation) $\quad \beta_U = \frac{\beta_E}{1+\frac{D}{E}(1-\tau_C)}$

Levered firm value $\quad V^L = V^U + PV(Interest\ tax\ shield)$

7 Corporate valuation

E, r_E	Value of equity and equity cost of capital
D, r_D	Value of debt and debt cost of capital
V	Firm value
r_U	Cost of capital of the unlevered firm
r_{WACC}	Cost of capital of the levered firm
τ_C	Marginal tax rate
$FCFE$	Expected Free Cash Flow to Equity
FCF	Expected Free Cash Flow to Firm

Equity method:
$$E = \sum_{t=1}^{\infty} FCFE_t \, (1 + r_E)^{-t}$$

Entity method:
$$V = \sum_{t=1}^{\infty} FCF_t \, (1 + r_{WACC})^{-t}$$

APV method:
$$V = \sum_{t=1}^{\infty} FCF_t \, (1 + r_U)^{-t} + \tau_C \sum_{t=1}^{\infty} D \cdot r_D (1 + r_D)^{-t}$$

$$V = \sum_{t=1}^{\infty} FCF_t \, (1 + r_U)^{-t} + \tau_C D$$